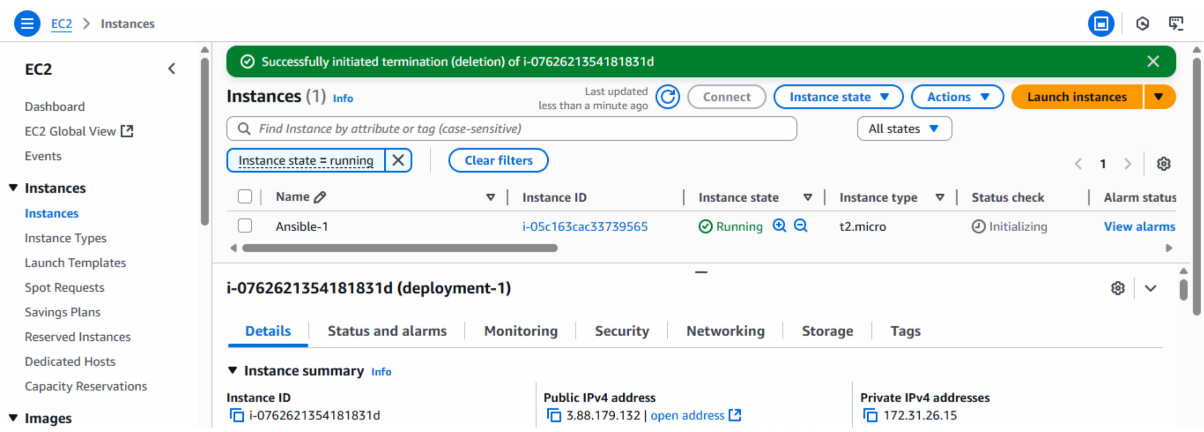
Ansible

Step:1 Create AWS Ubuntu Instance and try connecting it with aws cli



Save the .pem Key at some safe place

Step:2 Install Ansible in WSL

- sudo apt update
- sudo apt install ansible –y
- ansible –version

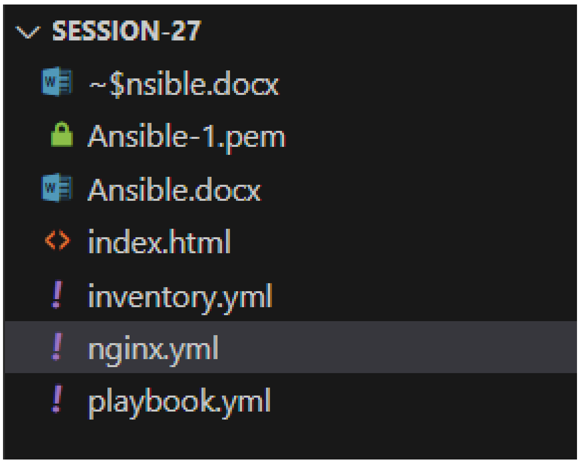
```
nikunj@DESKTOP-M4S6FPT:~$ ansible --version
ansible [core 2.16.3]
  config file = None
  configured module search path = ['/home/nikunj/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/nikunj/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Jun 18 2025, 17:59:45) [GCC 13.3.0] (/usr/bin/python3)
  jinja version = 3.1.2
  libyaml = True
nikunj@DESKTOP-M4S6FPT:~$ |
```

Step:3 Create One Folder and move your .pem key to it and create playbook

playbook.yml (or playbook.yaml)

```yaml
---
- name: Test Ansible Playbook
  hosts: localhost
  tasks:
  - name: Print Simple Message
    debug:
      msg: "Hello Ansible! if Working"
```

once playbook is ready > open the terminal in the same folder



to Run this playbook:

- ansible-playbook playbook.yml

output:

```
TASK [Print Simple Message] ***************************************************************************
ok: [localhost] => {
    "msg": "Hello Ansible! if Working"
}

PLAY RECAP ********************************************************************************************
localhost                  : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ig
nored=0

nikunj@DESKTOP-M4S6FPT:/mnt/d/pw/Live Classes/Devops April/Session-27$
```

Lets create an Ansible Script for AWS automation

inventory.yml

```yaml
all:
  hosts:
    aws1:
      ansible_host: 54.234.14.104
      ansible_user: ec2-user
      ansible_ssh_private_key_file: Ansible-1.pem
```

nginx.yml

```yaml
---
- name: Install nginx server
  hosts: aws1
  become: yes
  tasks:
    - name: Install nginx
      yum:
        name: nginx
        state: present
```

```yaml
  - name: Start nginx
    service:
      name: nginx
      state: started
      enabled: yes
  - name: copy the index.html to default location
    copy:
      src: index.html
      dest: /usr/share/nginx/html/index.html
      owner: nginx
      group: nginx
      mode: '0644'
  - name: Restart nginx
    service:
      name: nginx
      state: restarted
```

Step:4 Check the status:

- sudo ansible all --list-host -i inventory.yml
  this command will give you the number of active host

Step:5 Run the Nginx Installation and Create index.html file

- sudo ansible-playbook -i inventory.yml nginx.yml

```
nikunj@DESKTOP-M4S6FPT:/mnt/d/pw/Live Classes/Devops April/Session-27$ sudo ansible-playbook -i inventory.y
ml  nginx.yml

PLAY [Install nginx server] ***************************************************************************

TASK [Gathering Facts] ********************************************************************************
The authenticity of host '54.234.14.104 (54.234.14.104)' can't be established.
ED25519 key fingerprint is SHA256:rSsEpa6CJ4YFbNNXdHCv+3ZaR/ERGudQNJPdiPYMjiE.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
                                                            Completions quota reached
```

```
https://docs.ansible.com/ansible-core/2.16/reference_appendices/interpreter_discovery.html for more
information.
ok: [aws1]

TASK [Install nginx] **********************************************************************************
changed: [aws1]

TASK [copy the index.html to default location] ********************************************************
changed: [aws1]

TASK [Start nginx] ************************************************************************************
changed: [aws1]

PLAY RECAP ********************************************************************************************
aws1                       : ok=4    changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ig
nored=0
```

Goto> AWS instance> copy public IP Address> goto> Browser> ip_add:80